A Comparative Study of NLP Techniques for Predicting Stock Market Movement from News Headlines

Sneha Naidu
210048420
MSc Data Science
Sneha.Naidu@city.ac.uk

1 Problem statement and motivation

Financial markets are influenced by numerous factors including economic indicators, geopolitical events, and investor sentiment. Among these, news headlines provide one of the most immediate signals of market perception. However, the volume of daily news makes manual analysis impractical.

This project investigates whether natural language processing (NLP) can be used to predict stock market movement from news headlines. Specifically, the task is framed as a binary classification problem: given daily news headlines, predict whether the Dow Jones Industrial Average (DJIA) will increase (1) or decrease/remain unchanged (0).

The challenge lies in the noisy and ambiguous nature of financial news. Headlines often contain cautious or conflicting language, and the temporal structure of the data prevents random shuffling. This requires careful experimental design to avoid data leakage.

The core research question is:

> How do traditional and modern NLP techniques compare in predicting stock market movement from news headlines, and what does this reveal about feature representation in financial text?

2 Research hypothesis

This study hypothesises that while traditional NLP approaches such as bag-of-words and TF-IDF provide strong and interpretable baselines, domain-specific transformer models such as FinBERT will ultimately achieve superior performance due to their ability to capture contextual and semantic relationships within financial text.

Financial news headlines often contain domain-specific terminology and nuanced language where meaning depends heavily on context, negation, and phrasing. Traditional approaches treat text as a collection of independent tokens and therefore may fail to capture these subtleties. In contrast, transformer-based models leverage contextual embeddings, allowing them to interpret meaning based on surrounding words and broader sentence structure.

At the same time, it is expected that TF-IDF-based representations will remain competitive due to the highly structured and vocabulary-driven nature of financial news. Words such as “surge,” “fall,” or “beat expectations” carry strong directional signals, which may be effectively captured through term-weighting schemes without requiring deep contextual understanding.

Additionally, the choice of classifier is expected to influence performance. Models such as Support Vector Machines and Naive Bayes may outperform logistic regression when applied to TF-IDF features due to their ability to better handle high-dimensional sparse data. Finally, the use of time-aware validation strategies such as TimeSeriesSplit is expected to provide a more realistic estimate of model generalisation, particularly in the presence of shifting market conditions over time.

3 Related Work and Background

3.1 Sentiment Analysis in Finance

Previous research has shown that textual sentiment can provide useful signals for predicting financial market behaviour. Bollen et al. (2011) demonstrated that public mood derived from social media correlates with stock market trends, while Ding et al. (2015) highlighted the importance of event-driven information in financial news for market prediction.

However, financial text differs from general sentiment data. Words such as “decline”, “volatile”, or “beat expectations” carry domain-specific meanings that may not align with standard sentiment interpretations. As a result, general-purpose sentiment models often perform poorly in financial contexts. This limitation has led to the development of domain-specific models such as FinBERT (Araci, 2019), which are trained on financial corpora to better capture specialised language patterns.

3.2 Traditional vs Modern NLP Approaches

Traditional NLP methods such as Bag-of-Words (BoW) and TF-IDF represent text using frequency-based statistics. These approaches are computationally efficient, interpretable, and widely used in text classification tasks (Sebastiani, 2002). However, they treat words as independent features and ignore word order and contextual relationships.

To address these limitations, word embedding methods such as Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014) were introduced, allowing semantic relationships between words to be captured in vector space. Despite this improvement, these embeddings remain context-independent.

Transformer-based models such as BERT (Devlin et al., 2019) overcome this limitation by learning contextual embeddings that depend on surrounding words. FinBERT extends this approach by incorporating financial domain knowledge, resulting in improved performance on financial text tasks.

3.3 Evaluation for Time-Series Data

A key consideration in financial prediction is the temporal nature of the data. Standard cross-validation techniques can lead to data leakage by allowing future information to influence past predictions. As highlighted by Bergmeir and Benítez (2012), time-series models require evaluation strategies that respect temporal order.

To address this, time-aware methods such as TimeSeriesSplit are used, ensuring that models are trained on past data and evaluated on future data. This provides a more realistic estimate of model performance in real-world financial applications.

3.4 Research Gap

While existing studies demonstrate the effectiveness of both traditional and deep learning approaches for financial text analysis (Bollen et al. (2011); Ding et al. (2015); (Araci, 2019)), relatively few works provide a direct comparison between these methods under a controlled, time-aware evaluation framework.

In particular, it remains unclear whether the additional complexity of transformer-based models consistently leads to improved performance on smaller, noisy financial datasets. This project addresses this gap by systematically comparing Bag-of-Words, TF-IDF-based models, and a domain-specific transformer (FinBERT) using a consistent experimental setup and temporal validation strategy.

3.5 Accomplishments

- Task 1: Conduct literature review and define research problem – Completed
- Task 2: Preprocess and explore the dataset – Completed
- Task 3: Build and evaluate baseline model (Bag-of-Words + Logistic Regression) – Completed
- Task 4: Implement and compare TF-IDF-based models (Logistic Regression, SVM, Naive Bayes) – Completed
- Task 5: Implement and evaluate transformer-based model (FinBERT) – Completed
- Task 6: Explore word embedding models (Word2Vec/GloVe) – Not completed; replaced with FinBERT to prioritise a more advanced, domain-specific approach within time constraints
- Task 7: Perform cross-validation using TimeSeriesSplit – Completed
- Task 8: Conduct error analysis – Completed
- Task 9: Generate visualisations and final report – Completed

4 Approach and Methodology

This project implements a supervised learning pipeline designed to evaluate how different NLP techniques perform on financial news data. The pipeline consists of data preprocessing, feature representation, model training, and evaluation, with particular attention paid to avoiding temporal data leakage.

4.1 Data Pipeline and Preprocessing

The dataset consists of daily news headlines spanning from 2008 to 2016, where each day contains approximately 25 headlines. These headlines are concatenated into a single document per day to form the input text. The prediction target is a binary label indicating whether the stock market increased (1) or decreased/remained unchanged (0) on that day.

Standard preprocessing steps were applied to ensure consistency and reduce noise in the text data. These include lowercasing all text, removing punctuation and non-alphanumeric characters, and tokenizing the text into individual words. In some experimental settings, stopword removal was also applied to evaluate its impact on performance.

A key design decision in this project is the use of a chronological train-test split, where data from 2008–2014 is used for training and 2015–2016 for testing. This ensures that models are evaluated on future data only, preventing information leakage and better reflecting real-world deployment conditions.

4.2 Models and Text Representations

To provide a comprehensive comparison, five models were implemented, representing increasing levels of complexity in both feature representation and learning capability.

The simplest approach uses a bag-of-words representation combined with logistic regression, where text is represented as raw word counts. This serves as a baseline model, allowing us to evaluate whether more sophisticated methods provide meaningful improvements.

Building on this, TF-IDF representations are introduced to account for the relative importance of words across documents. Unlike bag-of-words, TF-IDF downweights common terms and emphasises discriminative words such as "crash" or "surge."

For TF-IDF representations, unigram and bigram features were used to capture both individual terms and short phrases, with features limited to the most informative terms to reduce noise. A linear kernel was used for the SVM model due to its effectiveness in high-dimensional text classification tasks. The Naive Bayes classifier was implemented as Multinomial Naive Bayes, which is well-suited for discrete word count features. Default hyperparameters were used to ensure fair and consistent comparison across models, as the primary objective of this study is to evaluate the impact of feature representation and model choice rather than extensive hyperparameter optimisation.

Finally, a transformer-based model, FinBERT, was fine-tuned on the dataset. FinBERT is pre-trained on financial corpora and is designed to capture contextual relationships within financial language. The model was trained using a weighted loss function to account for class imbalance, and decision thresholds were tuned to optimise F1-score.

4.3 Cross-Validation Strategy

Given the temporal nature of the dataset, traditional random cross-validation is inappropriate as it would introduce data leakage by allowing future information to influence past predictions. To address this, this study employs TimeSeriesSplit, a forward-chaining validation strategy where training data always precedes test data.

In this approach, the dataset is split into multiple folds such that each fold trains on earlier time periods and tests on subsequent periods. This simulates a realistic scenario in which models are continuously updated with historical data and evaluated on unseen future data. This strategy provides a more reliable estimate of model generalisation, particularly in financial contexts where market conditions evolve over time.

For the FinBERT model, due to computational constraints, a single chronological split is used alongside threshold tuning on the training data.

4.4 Evaluation Metrics

Model performance is evaluated using accuracy, precision, recall, and F1-score, with F1-score selected as the primary metric. While the dataset is relatively balanced, accuracy alone is not sufficient to distinguish between models, as all models achieve results close to random baseline performance.

More importantly, financial prediction involves asymmetric error costs. False positives (predicting market increases incorrectly) can lead to direct financial losses, whereas false negatives represent missed opportunities. F1-score provides a balanced measure that captures both precision and recall, making it more suitable for evaluating models in this context.

5 Dataset

The dataset used in this study is the *Combined News DJIA* dataset available on Kaggle (Sun, 2016), which links global news headlines with daily movements of the Dow Jones Industrial Average (DJIA). Each instance corresponds to a single trading day and contains up to 25 news headlines, along with a binary label indicating whether the market increased (1) or remained the same/decreased (0).

The dataset spans from August 2008 to July 2016 and contains 4,835 daily observations, representing approximately 120,000 headlines in total. On average, each day includes around 25 headlines. The class distribution is relatively balanced, with 51% of instances labelled as market increases and 49% as decreases or no change. This balance reduces the need for aggressive resampling, although class weighting was still considered for certain models.

Although the overall class distribution is balanced, the proportion of market movements varies over time due to changing financial conditions. This highlights the importance of time-aware evaluation, as models trained on earlier periods may not generalise well to later data. Additionally, many headlines contain mixed or weak sentiment signals, reflecting the noisy nature of the task.

To better understand the dataset, a word cloud of the most frequent terms across all headlines was generated (Figure 1). This provides a high-level overview of dominant vocabulary present in financial news.

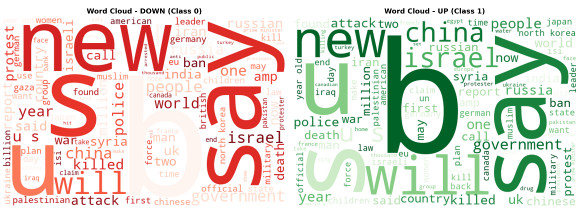

Figure 1: Word cloud

5.1 Dataset Characteristics and Challenges

Despite its suitability, the dataset presents several challenges. First, each day contains multiple headlines that may express conflicting signals, requiring the model to aggregate mixed sentiment into a single prediction. Second, financial news is inherently noisy and often uses cautious or speculative language (e.g., “may”, “could”), which weakens explicit sentiment signals.

Additionally, market movements are influenced by factors beyond news, such as technical indicators and macroeconomic trends, meaning that headlines alone may provide incomplete information. Finally, the temporal nature of the dataset introduces distribution shifts across different market regimes (e.g., post-2008 recovery vs. later periods), making generalisation more difficult.

5.2 Preprocessing

All headlines for a given day were concatenated into a single document to form the input text. This allows the model to learn from the combined information available within each trading day, rather than treating headlines independently.

Standard preprocessing steps such as lowercasing, punctuation removal, and tokenisation were applied to ensure consistency.

Stopword removal was treated as an experimental variable rather than a fixed step. While commonly used to reduce noise, removing stopwords can eliminate important linguistic cues such as negation. Therefore, its impact on performance was evaluated empirically.

6 Baselines

To ensure a fair and structured comparison, a set of baseline models was selected ranging from simple statistical methods to more advanced contextual models.

The Bag-of-Words (BoW) with Logistic Regression serves as a fundamental baseline, providing a reference point for evaluating improvements from more sophisticated representations. TF-IDF-based models extend this by incorporating term importance, allowing assessment of whether weighting improves performance over raw frequency counts.

Multiple classifiers (Logistic Regression, SVM, and Naive Bayes) were applied to TF-IDF features to evaluate how different learning paradigms perform under the same representation. Finally, FinBERT was included as a modern transformer-based model to compare traditional approaches with deep contextual methods.

This baseline design enables a controlled comparison across increasing levels of model complexity while maintaining consistency in evaluation.

7 Results and Analysis

This section presents the experimental results and analysis of all models introduced in Section 4. The objective is to compare how different text representations and model architectures perform in predicting stock market movement from news headlines.

7.1 Model Performance

Model performance was evaluated using accuracy, precision, recall, and F1-score, with F1-score treated as the primary metric due to its ability to capture the trade-off between precision and recall. This is particularly important in financial prediction, where false positives (incorrect buy signals) may lead to direct financial losses.

Table 1 presents the performance of all models on the test set, allowing direct comparison across different representations and classifiers.

Model	Accuracy	Precision	Recall	F1-score
BoW + LogReg	0.468	0.522	0.500	0.511
TF-IDF + LogReg	0.492	0.545	0.514	0.529
TF-IDF + SVM	0.524	0.583	0.500	0.538
TF-IDF + NB	0.516	0.541	0.857	**0.663**
FinBERT	0.532	0.568	0.657	0.609

Table 1: Model performance comparison

Table 1 shows a clear progression in performance as feature representation and model complexity increase. The Bag-of-Words baseline performs worst across all metrics, indicating that raw word frequency is insufficient for capturing meaningful financial signals. Introducing TF-IDF leads to consistent improvements, demonstrating the importance of weighting informative terms. Among TF-IDF models, Naive Bayes achieves the highest F1-score due to its significantly higher recall, while SVM and Logistic Regression provide more balanced but lower overall performance. Although FinBERT achieves the highest accuracy, its F1-score remains lower than Naive Bayes, suggesting that it does not capture positive market movements as effectively in this setting.

As shown in Figure 2, TF-IDF-based models consistently outperform the Bag-of-Words baseline across all evaluation metrics. In particular, the TF-IDF + Naive Bayes model achieves the highest F1-score, driven by its substantially higher recall compared to Logistic Regression, SVM, and FinBERT.

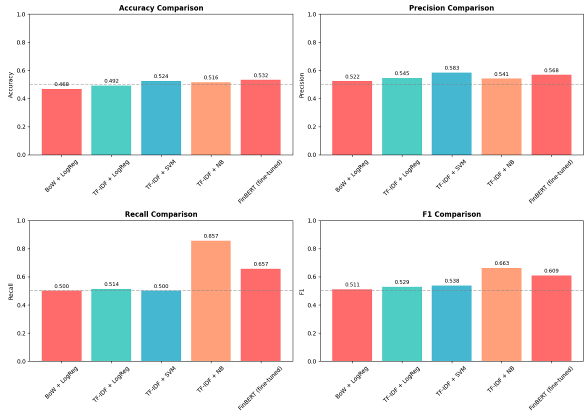

Figure 2: Model performance comparison across all models

These results highlight an important finding: although all models achieve similar accuracy levels close to random baseline, accuracy alone is not a reliable indicator of performance in this task. In contrast, F1-score reveals meaningful differences between models that are not captured by accuracy alone.

In this study, TF-IDF + Naive Bayes emerges as the strongest model, not because it achieves the highest accuracy, but because it is most effective at capturing positive market movements through its high recall. This indicates that the model achieves strong performance by capturing a large proportion of positive market movements (high recall), but at the cost of increased false positives, reflecting a trade-off between sensitivity and precision. The suitability of this trade-off depends on the specific financial strategy.

To better understand this behaviour, the confusion matrix of the TF-IDF + Naive Bayes model is shown in Figure 3.

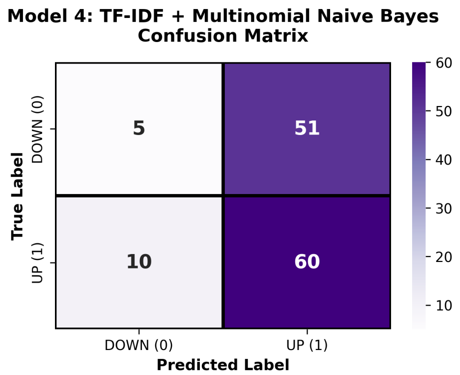

Figure 3: Confusion matrix for TF-IDF + Naive Bayes model

Although TF-IDF + Naive Bayes achieves the highest F1-score, Figure 3 reveals an important limitation of this model. The confusion matrix shows that while the model correctly identifies a large proportion of positive cases (high true positives), it also produces a significant number of false positives, indicating weaker performance in identifying negative cases.

Despite this limitation, Naive Bayes achieves the highest F1-score overall, indicating the best balance between precision and recall under the chosen evaluation metric. This highlights an important trade-off: the model is highly effective at capturing market increases but less reliable in avoiding incorrect positive predictions, which is particularly relevant in financial settings where false positives may lead to losses.

FinBERT achieved the highest accuracy (0.532) but a lower F1-score compared to Naive Bayes. This indicates that while FinBERT produces more balanced predictions with fewer false positives, it is less effective at capturing positive market movements. A possible explanation is that transformer models require larger datasets to fully exploit contextual information, whereas simpler models benefit from explicit keyword-based signals present in financial headlines.

Furthermore, both SVM and Logistic Regression, despite using the same TF-IDF features, underperform compared to Naive Bayes. This suggests that the probabilistic nature of Naive Bayes is particularly well-suited to capturing word distributions in financial text.

Additionally, TimeSeriesSplit cross-validation results show that TF-IDF + Naive Bayes achieves a mean F1-score of approximately 0.675 with low variance across folds, indicating consistent performance over time.

7.2 Impact of Feature Engineering

The experiments also highlight the importance of preprocessing choices. Incorporating bigram features provided a modest improvement over unigram representations, suggesting that short phrases such as “market surge” carry useful information. In contrast, removing stopwords significantly reduced performance, indicating that function words contribute to meaning, particularly in cases involving negation.

To further understand which terms drive predictions, the most important features from the TF-IDF + Naive Bayes model are shown in Figure 4.

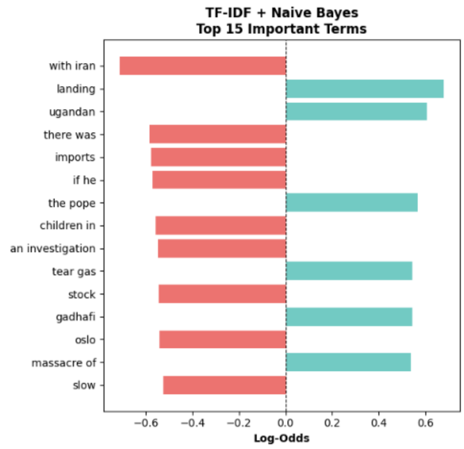

Figure 4: Top positive and negative words identified by TF-IDF + Naive Bayes

The figure shows that words such as “surge”, “gain”, and “record” are strongly associated with positive market movement, while words such as “fall”, “crash”, and “decline” are strong indicators of negative movement. This confirms that financial headlines often contain explicit keywords that can be effectively captured using statistical feature representations.

Overall, the results of this study show that TF-IDF + Naive Bayes is the most effective model for this task, achieving the highest F1-score by successfully capturing the strong keyword-driven patterns present in financial headlines. While more complex models such as FinBERT were expected to perform better, they did not provide a clear advantage on this dataset. This suggests that, in this case, model performance is driven more by the quality of feature representation than by model complexity, particularly given the relatively small dataset and the vocabulary-driven nature of financial text.

7.3 Error Analysis

To better understand model limitations, a manual analysis of approximately 25 misclassified examples was conducted. Several consistent error patterns were identified:

1. Mixed or Contradictory Signals (approximately 40%)

Examples often contained both positive and negative information within the same day, making prediction difficult.

Example: "Markets rise on strong earnings but investors remain cautious"

In such cases, models struggle to aggregate conflicting signals, often over-weighting one sentiment over the other.

2. Ambiguous or Neutral Language (approximately 25%)

Headlines with weak or unclear sentiment frequently led to incorrect predictions.

Example: "Investors await Fed decision amid uncertainty"

These headlines provide little directional signal, causing models to default to majority patterns.

3. Implicit Context and Market Dynamics (approximately 20%)

In some cases, the sentiment expressed in headlines did not align with actual market movement.

Example: "Tech stocks decline despite strong global outlook"

This highlights a key limitation: models rely only on textual input and cannot capture broader market context or external factors.

4. Temporal Effects (approximately 15%)

Market reactions do not always align with same-day news. Information may already be priced in or affect markets with a delay.

Overall, these patterns suggest that the primary challenge lies in the nature of the data rather than the models themselves. Financial headlines are often noisy, ambiguous, and incomplete, limiting the predictive power of any text-based approach.

8 Lessons Learned and Conclusions

Contrary to the initial hypothesis, the transformer-based model (FinBERT) did not outperform traditional approaches. Instead, TF-IDF combined with Naive Bayes achieved the highest F1-score, suggesting that simpler feature-based methods are more effective for this task. This result is likely due to the relatively small dataset and the strongly keyword-driven nature of financial headlines, where explicit terms such as "surge" or "decline" provide clear predictive signals that can be effectively captured without complex contextual modelling.

An important lesson from this project is that the relative importance of feature representation and model complexity is highly dependent on the characteristics of the data. In this case, where the dataset is relatively small and financial headlines contain strong keyword-driven signals, simpler statistical models outperform more complex approaches. However, for larger datasets or tasks requiring deeper contextual understanding, transformer-based models may provide greater benefits.

However, the study has several limitations. The dataset is relatively small, which restricts the ability of more complex models to fully learn contextual patterns. In addition, the labels are inherently noisy, as market movements are influenced by factors beyond news headlines. Furthermore, relying solely on textual data without incorporating price-based or macroeconomic features limits the overall predictive performance.

Overall, this project demonstrates that effective model selection must be guided by the characteristics of the data rather than assumptions about model sophistication. It also provides practical insight into the challenges of applying NLP techniques to real-world financial data, particularly the need to align model choice with data properties and evaluation objectives.

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